

1 Building the plug-in dll

The plug-ins for automatic layout and sequence in Tribon Plate Nesting are built in Microsoft Visual Studio as dynamically linked libraries with a define file. In the project described in this manual all interface implementation files are included through a library file. The implementation files may be included directly into the project. Below are all necessary files for the Visual Studio project and the interface included.

TbLayoutPlugin.sln

```
Microsoft Visual Studio Solution File, Format Version 8.00
Project("{8BC9CEB8-8B4A-11D0-8D11-00A0C91BC942}") =
    "TbLayoutPlugin", "TbLayoutPlugin.vcproj", "{9F4A9FD1-4AA1-
4D14-A025-14A65E001FDF}"
    ProjectSection(ProjectDependencies) = postProject
    EndProjectSection
EndProject
Global
    GlobalSection(SolutionConfiguration) = preSolution
        Debug = Debug
        Release = Release
    EndGlobalSection
    GlobalSection(ProjectConfiguration) = postSolution
        {9F4A9FD1-4AA1-4D14-A025-
14A65E001FDF}.Debug.ActiveCfg = Debug|Win32
        {9F4A9FD1-4AA1-4D14-A025-
14A65E001FDF}.Debug.Build.0 = Debug|Win32
        {9F4A9FD1-4AA1-4D14-A025-
14A65E001FDF}.Release.ActiveCfg = Release|Win32
        {9F4A9FD1-4AA1-4D14-A025-
14A65E001FDF}.Release.Build.0 = Release|Win32
    EndGlobalSection
    GlobalSection(ExtensibilityGlobals) = postSolution
    EndGlobalSection
    GlobalSection(ExtensibilityAddIns) = postSolution
    EndGlobalSection
EndGlobal
```

TbLayoutPlugin.vcproj

```
<?xml version="1.0" encoding="Windows-1252"?>
<VisualStudioProject
    ProjectType="Visual C++"
    Version="7.10"
    Name="TbLayoutPlugin"
    RootNamespace="TbLayoutPlugin"
    <Platforms>
        <Platform
```

```

        Name="Win32"/>
    </Platforms>
    <Configurations>
        <Configuration
            Name="Debug|Win32"
            OutputDirectory=".\\Debug"
            IntermediateDirectory=".\\Debug"
            ConfigurationType="2"
            UseOfMFC="0"

            ATLMinimizesCRunTimeLibraryUsage="FALSE"
            CharacterSet="2">
                <Tool

                    Name="VCCLCompilerTool"

                        Optimization="0"

                        PreprocessorDefinitions="WIN32;_DEBUG;_WINDOWS;_USR
DLL;TBLAYOUTPLUGIN_EXPORTS;_WIN32_WINNT=0x0500;WINVER=0x0500"
                        BasicRuntimeChecks="3"
                        RuntimeLibrary="1"

                        PrecompiledHeaderFile=".\\Debug\\TLayoutPlugin.pch"

                        AssemblerListingLocation=".\\Debug/"
                        ObjectFile=".\\Debug/"

                        ProgramDataBaseFileName=".\\Debug/"
                        WarningLevel="3"

                        SuppressStartupBanner="TRUE"

                        DebugInformationFormat="4"/>
                <Tool

                    Name="VCCustomBuildTool"/>
                <Tool

                    Name="VCLinkerTool"

                        AdditionalDependencies="kcsanstl_d.lib odbcc32.lib
odbccp32.lib"

                        OutputFile=".\\Debug\\TLayoutPlugin.dll"
                        LinkIncremental="1"

                        SuppressStartupBanner="TRUE"

                        AdditionalLibraryDirectories="&quot;<SPECIFY
ADDITIONAL LIBRARY DIRECTORIES HERE>&quot;;"

                        IgnoreDefaultLibraryNames="libc.lib, libcmtd.lib, msvc
rt.lib, libcd.lib, msvcrtd.lib"

```

```
ModuleDefinitionFile=".\\TBLayoutPlugin.def"

GenerateDebugInformation="TRUE"

ProgramDatabaseFile=".\\Debug/TBLayoutPlugin.pdb"

ImportLibrary=".\\Debug/TBLayoutPlugin.lib"
                                TargetMachine="1"/>
    <Tool
                                Name="VCIDLTool"

PreprocessorDefinitions="_DEBUG"

MkTypLibCompatible="TRUE"

SuppressStartupBanner="TRUE"
                                TargetEnvironment="1"

TypeLibraryName=".\\Debug/TBLayoutPlugin.tlb"
                                HeaderFileName=""/>
    <Tool

    Name="VCPostBuildEventTool"
                                CommandLine="call
%MAKE_TOOLS_CC%\\xinstall -c $(TargetPath)
%PDMSINSTALL%\\$(ConfigurationName)"/>
    <Tool

    Name="VCPreBuildEventTool"/>
    <Tool

    Name="VCPreLinkEventTool"/>
    <Tool

    Name="VCResourceCompilerTool"

PreprocessorDefinitions="_DEBUG"
                                Culture="1033"

AdditionalIncludeDirectories=""/>
    <Tool

    Name="VCWebServiceProxyGeneratorTool"/>
    <Tool

    Name="VCXMLDataGeneratorTool"/>
    <Tool

    Name="VCWebDeploymentTool"/>
    <Tool

    Name="VCManagedWrapperGeneratorTool"/>
```

```

        <Tool
          Name="VCAuxiliaryManagedWrapperGeneratorTool"/>
        </Configuration>
        <Configuration
          Name="Release|Win32"
          OutputDirectory=".\\Release"

          IntermediateDirectory=".\\Release"
          ConfigurationType="2"
          UseOfMFC="0"

          ATLMinimizesCRunTimeLibraryUsage="FALSE"
          CharacterSet="2">
        <Tool

          Name="VCCLCompilerTool"

          Optimization="2"

          InlineFunctionExpansion="1"

          PreprocessorDefinitions="WIN32;NDEBUG;_WINDOWS;_USR
DLL;TBLAYOUTPLUGIN_EXPORTS;_WIN32_WINNT=0x0500;WINVER=0x0500"
          StringPooling="TRUE"
          RuntimeLibrary="0"

          EnableFunctionLevelLinking="TRUE"

          PrecompiledHeaderFile=".\\Release/TBLayoutPlugin.pch
"

          AssemblerListingLocation=".\\Release/"

          ObjectFile=".\\Release/"

          ProgramDataBaseFileName=".\\Release/"
          WarningLevel="3"

          SuppressStartupBanner="TRUE"/>
        <Tool

          Name="VCCustomBuildTool"/>
        <Tool

          Name="VCLinkerTool"

          AdditionalDependencies="kcsanstl.lib odbcc32.lib
odbccp32.lib"

          OutputFile=".\\Release/TBLayoutPlugin.dll"
          LinkIncremental="1"

          SuppressStartupBanner="TRUE"

```

```
        AdditionalLibraryDirectories="<SPECIFY ADDITIONAL
LIBRARY DIRECTORIES HERE>"

        IgnoreDefaultLibraryNames="libc.lib, libcmtd.lib, libcd.lib, libcmtd.lib, msvcrt.lib"

        ModuleDefinitionFile=".\\TBLayoutPlugin.def"

        ProgramDatabaseFile=".\\Release/TBLayoutPlugin.pdb"

        ImportLibrary=".\\Release/TBLayoutPlugin.lib"
                                TargetMachine="1"/>
        <Tool
                                Name="VCIDLTool"

        PreprocessorDefinitions="NDEBUG"

        MkTypLibCompatible="TRUE"

        SuppressStartupBanner="TRUE"                                TargetEnvironment="1"

        TypeLibraryName=".\\Release/TBLayoutPlugin.tlb"
                                HeaderFileName=""/>
        <Tool

        Name="VCPostBuildEventTool"                                CommandLine="call
%MAKE_TOOLS_CC%\\xinstall -c $(TargetPath)
%PDMSINSTALL%\\$(ConfigurationName)
call %CM_BIN%\\move_file.bat $(TargetDir) $(TargetFileName)
bin"/>
        <Tool

        Name="VCPreBuildEventTool"/>
        <Tool

        Name="VCPreLinkEventTool"/>
        <Tool

        Name="VCResourceCompilerTool"

        PreprocessorDefinitions="NDEBUG"                                Culture="1033"

        AdditionalIncludeDirectories=""/>
        <Tool

        Name="VCWebServiceProxyGeneratorTool"/>
        <Tool

        Name="VCXMLDataGeneratorTool"/>
```

```

        <Tool
            Name="VCWebDeploymentTool"/>
        <Tool
            Name="VCManagedWrapperGeneratorTool"/>
        <Tool
            Name="VCAuxiliaryManagedWrapperGeneratorTool"/>
        </Configuration>
    </Configurations>
    <References>
    </References>
    <Files>
        <Filter
            Name="Source Files"
            Filter="cpp;c;cxx;rc;def;r;odl;idl;hpj;bat;f90;for;
f;fpp">
            <File
                RelativePath="TBLayoutPlugin.cpp">
                    <FileConfiguration
                        Name="Debug|Win32">
                            <Tool
                                Name="VCCLCompilerTool"
                                Optimization="0"
                                PreprocessorDefinitions="WIN32;_DEBUG;_WINDOWS;_MBC
S;_USRDLL;TBLAYOUTPLUGIN_EXPORTS;_WIN32_WINNT=0x0500;WINVER=0
x0500;$(NoInherit)"
                                BasicRuntimeChecks="3"/>
                            </FileConfiguration>
                        <FileConfiguration
                            Name="Release|Win32">
                                <Tool
                                    Name="VCCLCompilerTool"
                                    Optimization="2"
                                    PreprocessorDefinitions="WIN32;NDEBUG;_WINDOWS;_MBC
S;_USRDLL;TBLAYOUTPLUGIN_EXPORTS;$(NoInherit)"/>
                                </FileConfiguration>
                            </File>
                        <File
                            RelativePath="TBLayoutPlugin.def">

```

```

        </File>
    </Filter>
    <Filter
        Name="Header Files"
        Filter="h;hpp;hxx;hm;inl;fi;fd">
        <File

RelativePath="TBLayOutPlugin.h">
            </File>
        </Filter>
        <Filter
            Name="Resource Files"

Filter="ico;cur;bmp;dlg;rc2;rct;bin;rgs;gif;jpg;jpe
g;jpe">
            <File

RelativePath="resource.h">
                </File>
            </Filter>
        </Files>
    <Globals>
    </Globals>
</VisualStudioProject>

```

TBLayoutPlugin.h

```
// tblayoutplugin.h: interface for the tblayoutplugin class.
//
////////////////////////////////////
////////

#ifdef
!defined(AFX_TBLAYOUTPLUGIN_H__259152F2_03E8_11D5_BD54_080009
0A4C97__INCLUDED_)
#define
AFX_TBLAYOUTPLUGIN_H__259152F2_03E8_11D5_BD54_0800090A4C97__I
NCLUDED_

#ifdef TBLAYOUTPLUGIN_EXPORTS
#define TBLAYOUTPLUGIN_API __declspec(dllexport)
#else
#define TBLAYOUTPLUGIN_API __declspec(dllimport)
#endif

#ifdef _MSC_VER > 1000
#pragma once
#endif // _MSC_VER > 1000

TBLAYOUTPLUGIN_API int wo001(void);

TBLAYOUTPLUGIN_API void wo002 ( const int NumberOfRawPlates);

TBLAYOUTPLUGIN_API void wo003( const int RawPlateNo,
                                const int NumberOfSegParts,
                                const char * Quality,
                                const int QuantityTotal,
                                const int QuantityNormal,
                                const int QuantityMirror,
                                const double Thickness,
                                const double PartThickMin,
                                const double PartThickMax,
                                const double DistPartPlate,
                                const double DistPartPart);

TBLAYOUTPLUGIN_API void wo004 ( const int RawPlateNo,
                                const int SegPart,
                                const int SegType,
                                const double AmpU,
                                const double AmpV,
                                const double EndU,
                                const double EndV);

TBLAYOUTPLUGIN_API void wo005 ( const int
                                NumberOfPlateParts);

TBLAYOUTPLUGIN_API void wo006 ( const int PlatePartNo,
```



```
const int ContourNo,
const int NumberOfSegParts);

TBLAYOUTPLUGIN_API void wo007( const int PlatePartNo,
                                const int MirrorPartNo,
                                const int NumberOfContours,
                                const char * PartName,
                                const char * PositionNumber,
                                const char * Quality,
                                const double Thickness,
                                const int QuantityNormal,
                                const int QuantityMirror,
                                const double PartLength,
                                const double PartWidth,
                                const double PartAngle,
                                const double
                                    MinCircRectOrigin[2],
                                const double Cog[2],
                                const double Perimeter,
                                const double
                                    EffectivePartArea,
                                const double BoundaryPartArea,
                                const char * Assembly,
                                const char * ProductionDate,
                                const double RotationAngle,
                                const int MirrorFlag);

TBLAYOUTPLUGIN_API void wo008( const int PartNo,
                                const int ContourNo,
                                const int SegPart,
                                const int SegType,
                                const double AmpU,
                                const double AmpV,
                                const double EndU,
                                const double EndV);

TBLAYOUTPLUGIN_API void wo009 ( const int BaseQualityNumber,
                                const int NumberOfQualities);

TBLAYOUTPLUGIN_API void wo010 ( const int BaseQualityNumber,
                                const char * BaseQuality ,
                                const char * Quality );

TBLAYOUTPLUGIN_API int wo011 ( void);

TBLAYOUTPLUGIN_API int wo012 ( void);

TBLAYOUTPLUGIN_API void wo013( const int NestedPlateNo ,
                                int& RawPlateNo ,
                                int& QuantityNormal ,
                                int& QuantityMirror ,
                                char* Assembly ,
                                char* ProductionDate ,
```

```
int& NumberOfNestedParts );

TBLAYOUTPLUGIN_API void wo014( const int  NestedPlateNo ,
                                const int  NestedPartNo ,
                                int& PlatePartNo ,
                                int& Nested ,
                                double& Dx ,
                                double& Dy ,
                                double& Angle ,
                                int& Mirr );

TBLAYOUTPLUGIN_API void wo015 ( void);

TBLAYOUTPLUGIN_API void wo016 ( const int
                                NumberOfBaseQualities ,
                                const int
                                MaxNumberOfQualities );

TBLAYOUTPLUGIN_API void wo017 ( const int  NestSet ,
                                const int  PlateFill ,
                                const double MaxScrapSingle ,
                                const double MaxScrapMirror ,
                                const double MaxScrapMulti ,
                                const int  TypeOfOptimization );

TBLAYOUTPLUGIN_API void wo018( const int  NestedPlateNo ,
                                const int  RawPlateNo ,
                                const int  QuantityNormal ,
                                const int  QuantityMirror ,
                                const char * Assembly ,
                                const char * ProductionDate ,
                                const int
                                NumberOfNestedParts);

TBLAYOUTPLUGIN_API void wo019( const int  NestedPlateNo ,
                                const int  NestedPartNo ,
                                const int  PartNo ,
                                const double Dx ,
                                const double Dy ,
                                const double Angle ,
                                const int  Mirr );

TBLAYOUTPLUGIN_API int wo020 ( void);

TBLAYOUTPLUGIN_API void wo021( const int  RemainedPartNo ,
                                int& PartNo ,
                                int& QuantityNormal ,
                                int& QuantityMirror ,
                                int& ReasonFlag );

TBLAYOUTPLUGIN_API void wo022( const int  NestedPlateNo,
                                char* NestingName,
                                int& NestingType,
                                char* DrawingNumber);
```

```
TBLAYOUTPLUGIN_API void wo023( const int PlatePartNo,
                                const int ContourNo,
                                const int NumberOfBevels);

TBLAYOUTPLUGIN_API void wo024( const int PlatePartNo,
                                const int ContourNo,
                                const int BevelIntervalNo,
                                const int SegPartStart,
                                const double StartPoint[2],
                                const int SegPartEnd,
                                const double EndPoint[2],
                                const double BevelCode);

TBLAYOUTPLUGIN_API void wo025( const int NestedPlateNo,
                                double& Length,
                                double& Width,
                                double& Thickness,
                                char* Quality,
                                double& Density);

#endif //
!defined(AFX_TBLAYOUTPLUGIN_H__259152F2_03E8_11D5_BD54_080009
0A4C97__INCLUDED_)
```

TBLayoutPlugin.cpp

```
// TBLayoutPlugin.cpp : Defines the entry point for the DLL  
application.  
//
```

```
#include "windows.h"  
#include "TBLayoutPlugin.h"
```

2 Interface routines

WO001

NAME:

WO001 - AutoNest Interface routine - Is External AutoNest
Connected

PURPOSE:

The procedure checks if an external AutoNest algorithm has
been connected.

DECLARATION:

```
int wo001 ( void);
```

RESULT:

The procedure returns TRUE if connected, otherwise FALSE.

```
-----*/
/*
 *      External declarations
 */
extern "C"
{
#ifdef TBLAYOUTPLUGIN_API
#define TBLAYOUTPLUGIN_API
#endif

/*
*****
*
*   Main procedure - wo001
*
*****
*/
#ifdef WIN32
TBLAYOUTPLUGIN_API
#endif
    int wo001 ( void)
    {
        return 1;
    }
}
```

WO002

NAME:

WO002 - AutoNest Interface routine - Define Number of Raw Plates

PURPOSE:

The procedure defines the number of raw plates available in the autonest session.

DECLARATION:

```
void wo002 ( const int NumberOfRawPlates);
```

INPUT:

Parameters:

NumberOfRawPlates The number of available raw plates

RESULT:

None.

```
-----*/
/*
 *      External declarations
 */
extern "C"
{
#ifdef TBLAYOUTPLUGIN_API
#define TBLAYOUTPLUGIN_API
#endif

/*****
 *
 *   Main procedure - wo002
 *
 *****/
#ifdef WIN32
TBLAYOUTPLUGIN_API
#endif
    void wo002 ( const int /* NumberOfRawPlates */)
    {
    }

/*
 * Allocate C++ structure for the given number of raw plates. Data will
 * be added in the calls of wo003 and wo004.
 */
}
```

WO003

NAME:

WO003 - AutoNest Interface routine - Add Raw Plate

PURPOSE:

The procedure adds the data for one raw plate.

DECLARATION:

```
void wo003( const int RawPlateNo,
            const int NumberOfSegParts,
            const char *Quality,
            const int QuantityTotal,
            const int QuantityNormal,
            const int QuantityMirror,
            const int FlangeSide,
            const double Thickness,
            const double PartThickMin,
            const double PartThickMax,
            const double DistPartPlate,
            const double DistPartPart,
            const double DistPartPlate2,
            const double DistPartPart2);
```

INPUT:

Parameters:

RawPlateNo	The identification number for the raw plate
NumberOfSegParts	The number of segment parts in the raw plate geometry
Quality	The raw plate quality
QuantityTotal	The total number of identical nestings (QuantityTotal = QuantityNormal + QuantityMirror)
QuantityNormal	The number of nestings burned normal
QuantityMirror	The number of nestings burned mirrored
FlangeSide	Code for reflecting parts with folded flanges: = -1 Force folded flange to be at the other side = 0 Automatic = +1 Force folded flange to be at this side

Thickness	The raw plate thickness
PartThickMin	The minimum allowed plate part thickness that can be nested on the raw plate
PartThickMax	The maximum allowed plate part thickness that can be nested on the raw plate
DistPartPlate	The minimum distance between the raw plate and a part
DistPartPart	The minimum distance between a part and a part
DistPartPlate2	The minimum distance between the raw plate and a part (Nest type = 2)
DistPartPart2	The minimum distance between a part and a part (Nest type = 2)

RESULT:

```

        None.
-----*/
/*
 *      External declarations
 */
extern "C"
{
#ifdef TBLAYOUTPLUGIN_API
#define TBLAYOUTPLUGIN_API
#endif

/*
*****
*
*   Main procedure - wo003
*
*****
*/
#ifdef WIN32
TBLAYOUTPLUGIN_API
#endif
    void wo003( const int /* RawPlateNo */,
                const int /* NumberOfSegParts */,
                const char * /* Quality */,
                const int /* QuantityTotal */,
                const int /* QuantityNormal */,
                const int /* QuantityMirror */,
                const int /* FlangeSide */,
                const double /* Thickness */,
                const double /* PartThickMin */,
                const double /* PartThickMax */,

```



```
        const double /* DistPartPlate */,
        const double /* DistPartPart */,
        const double /* DistPartPlate2 */,
        const double /* DistPartPart2 */)
    {
/* Store the given data in the C++ structure from wo002.
 * Allocate C++ structure for the raw plate contour with the given
 * number of segment parts. The contour will be added in wo004.
 */
    }
}
```

W0004

NAME:

W0004 - AutoNest Interface routine - Add Segment Part for Raw Plate

PURPOSE:

The procedure adds one segment part for the raw plate contour. The first segment part consists only of the starting point. Each successive segment part contains the amplitude vector and the ending point. 2 consecutive segment parts constitutes one segment.

DECLARATION:

```
void wo004( const int RawPlateNo,
            const int SegPart,
            const int SegType,
            const double AmpU,
            const double AmpV,
            const double EndU,
            const double EndV);
```

INPUT:

Parameters:

RawPlateNo	The identification number for the raw plate
SegPart	The segment part number, starting from 0
SegType	The segment part type: 0 = The starting point 1 = Line 2 = Arc
AmpU AmpV	The amplitude vector
EndU EndV	The ending point

RESULT:

None.

-----*/

```
/*
 *      External declarations
 */
extern "C"
{
#ifdef TBLAYOUTPLUGIN_API
#define TBLAYOUTPLUGIN_API
#endif

/*
*****
*
*   Main procedure - wo004
*
*****
*/
#ifdef WIN32
TBLAYOUTPLUGIN_API
#endif
        void wo004( const int /* RawPlateNo */,
                    const int /* SegPart */,
                    const int /* SegType */,
                    const double /* AmpU */,
                    const double /* AmpV */,
                    const double /* EndU */,
                    const double /* EndV */)
        {
/* Store the given data in the C++ structure from wo003.
* Allocate C++ structure for the raw plate contour with the given
* number of segment parts. The contour will be added in wo004.
*/
                }
}
```

WO005**NAME:**

WO005 - AutoNest Interface routine - Define Number of Plate Parts

PURPOSE:

The procedure defines the number of available plate parts in the autonest session.

DECLARATION:

```
void wo005 ( const int NumberOfPlateParts);
```

INPUT:**Parameters:**

NumberOfPlateParts	The number of available plate parts
--------------------	-------------------------------------

RESULT:

None.

```
-----*/
/*
 *      External declarations
 */
extern "C"
{
#ifdef TBLAYOUTPLUGIN_API
#define TBLAYOUTPLUGIN_API
#endif

/*
*****
*
*   Main procedure - wo005
*
*****
*/
#ifdef WIN32
TBLAYOUTPLUGIN_API
#endif
    void wo005 ( const int /* NumberOfPlateParts */)
    {
    }
}
```

WO006**NAME:**

WO006 - AutoNest Interface routine - Define number of segment parts

PURPOSE:

The procedure defines the number of segment parts for one contour.

DECLARATION:

```
void wo006 ( const int PlatePartNo,  
             const int ContourNo,  
             const int NumberOfSegParts,  
             const int ContourType,  
             const int MarkingType,  
             const int MarkingSide);
```

INPUT:**Parameters:**

PlatePartNo	The identification number for the plate part
ContourNo	The contour number, starting at 1
NumberOfSegParts	The number of segment parts for the contour
ContourType	The type of contour: = 1 Outer contour = 2 Hole = 3 Marking = 4 Blasting = 5 GSD
MarkingType	The type of marking/blasting/GSD: = 0 Hull curve or no reference = 1 Hole = 2 Bracket = 3 Inner seam in PLCM = 4 Stiffener = 5 Flange = 6 Added contour = 7 Bracket notch = 8 Knuckle line = 9 Reference plane = 10 Pipe penetration = 11 Nominal contour = 12 Web with flange geometry

- = 13 Flange with web geometry
- = 14 Structure
- = 15 Swedging
- = 16 Metal touching line
- = 17 Small hole
- = 18 Bracket overlap
- = 19 Intersecting panel
- = 20 Intersecting bracket on another panel
- = 21 Cutout crossing
- = 22 Abutting profile
- = 23 Abutting shell profile
- = 24 Reference line in PLCM
- = 25 Attachment angle, intersecting panel
- = 26 Attachment angle, intersecting bracket
- = 27 Attachment angle, profile
- = 28 Attachment angle, bracket
- = 29 GSD at plate part corner
- = 30 GSD, manually added
- = 31 GSD at notches for seams/butts in the shell
- = 32 GSD at CL

MarkingSide

The side of marking/blasting/GSD:

- = +1 This side
- = 0 Both sides
- = -1 Other side

Irrelevant for burning contours.

RESULT:

None.

-----*/

```
/*
 *      External declarations
 */
extern "C"
{
#ifdef TBLAYOUTPLUGIN_API
#define TBLAYOUTPLUGIN_API
#endif

/*
*****
*
*   Main procedure - wo006
*
*****
*/
#ifdef WIN32
TBLAYOUTPLUGIN_API
#endif
    void wo006 ( const int /* PlatePartNo */,
                  const int /* ContourNo */,
                  const int /* NumberOfSegParts */,
                  const int /* ContourType */,
                  const int /* MarkingType */,
                  const int /* MarkingSide */)
    {
    }
}
```

WO007

NAME:

WO007 - AutoNest Interface routine - Add Plate Part

PURPOSE:

The procedure adds a plate part.

DECLARATION:

```
void wo007( const int PlatePartNo,
            const int MirrorPartNo,
            const int NumberOfContours,
            const char *PartName,
            const char *PositionNumber,
            const char *Quality,
            const double Thickness,
            const int QuantityNormal,
            const int QuantityMirror,
            const double PartLength,
            const double PartWidth,
            const double PartAngle,
            const double MinCircRectOrigin[2],
            const double Cog[2],
            const double Perimeter,
            const double EffectivePartArea,
            const double BoundaryPartArea,
            const char *Assid1,
            const char *Assid2,
            const char *Assid3,
            const char *Assid4,
            const char *Assembly,
            const char *ProductionDate,
            const double RotationAngle,
            const int MirrorFlag);
```

INPUT:

Parameters:

PlatePartNo	The identification number for the plate part
MirrorPartNo	The identification number for the plate part which is the mirrored image of the current plate part. The data for the symmetrical part will follow as the next part. The data for the mirrored part will be used in the single nesting of the part.
	This parameter is only given when symmetrical parts are stored as

	individual objects. Otherwise MirrPartNo = 0.
NumberOfContours	The number of contours
PartName	The plate part name
PositionNumber	The plate part position number
Quality	The plate part quality
Thickness	The plate part thickness
QuantityNormal	The number of normal parts
QuantityMirror	The number of mirrored parts
PartLength	The length of the part along the longer side of the least circumscribed rectangle
PartWidth	The width of the part along the shorter side of the least circumscribed rectangle
PartAngle	The angle between the X-axis and the longer side of the least circumscribed rectangle
MinCircRectOrigin	The origin of the least circumscribed rectangle
CoG	The centre of gravity of the plate part boundary
Perimeter	The perimeter of the plate part
EffectivePartArea	The plate part area with the area of the holes subtracted
BoundaryPartArea	The area of the plate part boundary
Assid1- Assid4	The assembly id's 1-4 from the General Purpose Strings (GPS)
Assembly	The plate part assembly
ProductionDate	The production date
RotationAngle	The rotation angle to be used in the nesting
MirrorFlag	The mirror flag: 0 = No mirroring allowed 1 = Mirroring allowed

RESULT:
None.

```

-----*/
/*
 *      External declarations
 */
extern "C"
{
#ifdef TBLAYOUTPLUGIN_API
#define TBLAYOUTPLUGIN_API
#endif

/*
 *
 *      Main procedure - wo007
 *
 *
 */
#ifdef WIN32
TBLAYOUTPLUGIN_API
#endif
    void wo007( const int /* PlatePartNo */,
                const int /* MirrorPartNo */,
                const int /* NumberOfContours */,
                const char * /* PartName */,
                const char * /* PositionNumber */,
                const char * /* Quality */,
                const double /* Thickness */,
                const int /* QuantityNormal */,
                const int /* QuantityMirror */,
                const double /* PartLength */,
                const double /* PartWidth */,
                const double /* PartAngle */,
                const double [2] /* MinCircRectOrigin */,
                const double [2] /* Cog */,
                const double /* Perimeter */,
                const double /* EffectivePartArea */,
                const double /* BoundaryPartArea */,
                const char * /* Assid1 */,
                const char * /* Assid2 */,
                const char * /* Assid3 */,
                const char * /* Assid4 */,
                const char * /* Assembly */,
                const char * /* ProductionDate */,
                const double /* RotationAngle */,
                const int /* MirrorFlag */)
    {
    }
}

```

WO008**NAME:**

WO008 - AutoNest Interface routine - Add Segment Part for Plate Part

PURPOSE:

The procedure adds one segment part for a plate part contour. The first segment part consists only of the starting point. Each successive segment part contains the amplitude vector and the ending point. 2 consecutive segment parts constitutes one segment.

DECLARATION:

```
void wo008( const int PartNo,
            const int ContourNo,
            const int SegPart,
            const int SegType,
            const double AmpU,
            const double AmpV,
            const double EndU,
            const double EndV);
```

INPUT:

Parameters:

PartNo	The identification number for the plate part
ContourNo	The plate part contour number, starting from
1	
SegPart	The segment part number, starting from 0
SegType	The segment part type: 0 = The starting point 1 = Line 2 = Arc
AmpU AmpV	The amplitude vector
EndU EndV	The ending point

RESULT:

None.

----- */

```
/*
 *      External declarations
 */
extern "C"
{
#ifdef TBLAYOUTPLUGIN_API
#define TBLAYOUTPLUGIN_API
#endif

/*
*****
*
*   Main procedure - wo008
*
*****
*/
#ifdef WIN32
TBLAYOUTPLUGIN_API
#endif
    void wo008( const int /* PartNo */,
                const int /* ContourNo */,
                const int /* SegPart */,
                const int /* SegType */,
                const double /* AmpU */,
                const double /* AmpV */,
                const double /* EndU */,
                const double /* EndV */)
    {
    }
}
```

WO009**NAME:**

WO009 - AutoNest Interface routine - Define Number of Qualities

PURPOSE:

The procedure defines the number of qualities for a base quality.

DECLARATION:

```
void wo009 ( const int BaseQualityNumber,  
             const int NumberOfQualities);
```

INPUT:**Parameters:**

BaseQualityNumber	The order number of the base quality, starting at 1
NumberOfQualities	The number of qualities for the give base quality

RESULT:

None.

-----*/

```
/*
 *      External declarations
 */
extern "C"
{
#ifdef TBLAYOUTPLUGIN_API
#define TBLAYOUTPLUGIN_API
#endif

/*
*****
*
*   Main procedure - wo009
*
*****
*/
#ifdef WIN32
TBLAYOUTPLUGIN_API
#endif
    void wo009 ( const int /* BaseQualityNumber */,
                  const int /* NumberOfQualities */)
    {
    }
}
```

WO010**NAME:**

WO010 - AutoNest Interface routine - Add Quality for Quality Exchange

PURPOSE:

The procedure adds one quality to the quality exchange.

DECLARATION:

```
void wo010 ( const int BaseQualityNumber,  
             const char *BaseQuality,  
             const char *Quality);
```

INPUT:**Parameters:**

BaseQualityNumber	The number of the current base quality
BaseQuality	The base quality
Quality	A quality on which the BaseQuality can be nested

RESULT:

None.

-----*/

```
/*
 *      External declarations
 */
extern "C"
{
#ifdef TBLAYOUTPLUGIN_API
#define TBLAYOUTPLUGIN_API
#endif

/*
*****
*
*   Main procedure - wo010
*
*****
*/
#ifdef WIN32
TBLAYOUTPLUGIN_API
#endif
    void wo010 ( const int /* BaseQualityNumber*/,
                  const char * /* BaseQuality */,
                  const char * /* Quality */)
    {
    }
}
```


WO011

NAME:

WO011 - AutoNest Interface routine - Autonest

PURPOSE:

The procedure starts the autonesting.

DECLARATION:

int wo011 (void);

INPUT:

RESULT:

The procedure returns 1 when the autonesting is finished,
otherwise 0.

```
-----*/
/*
 *      External declarations
 */
extern "C"
{
#ifdef TBLAYOUTPLUGIN_API
#define TBLAYOUTPLUGIN_API
#endif

/*
*****
*
*   Main procedure - wo011
*
*****
*/
#ifdef WIN32
TBLAYOUTPLUGIN_API
#endif
    int wo011 ( void)
    {
        return 0;
    }
}
```

WO012

NAME:

WO012 - AutoNest Interface routine - Get Number of Nested
Plates

PURPOSE:

The procedure gets the number of nested plates.

DECLARATION:

int wo012(void);

INPUT:

RESULT:

The number of nested plates is returned.

```
----- */
/*
 *      External declarations
 */
extern "C"
{
#ifdef TBLAYOUTPLUGIN_API
#define TBLAYOUTPLUGIN_API
#endif

/*
*****
*
*   Main procedure - wo012
*
*****
*/
#ifdef WIN32
TBLAYOUTPLUGIN_API
#endif
    int wo012( void)
    {
        return 0;
    }
}
```

NAME:

W0013 - AutoNest Interface routine - Get Number of Nested Parts

PURPOSE:

The procedure gets the number of nested parts for the given nested plate.

DECLARATION:

```
void wo013( const int NestedPlateNo,
            int& RawPlateNo,
            int& QuantityNormal,
            int& QuantityMirror,
            char* Assembly,
            char* ProductionDate,
            int& NumberOfNestedParts);
```

INPUT:

Parameters:

NestedPlateNo	The identification number for the nested plate
---------------	--

RESULT:

Parameters:

RawPlateNo	The identification number for the used raw plate
QuantityNormal	The number of nestings burned normal
QuantityMirror	The number of nestings burned mirrored
Assembly	Assembly for most of the parts
ProductionDate	Production date for most of the parts
NumberOfNestedParts	The number of nested parts

```
-----*/
/*
 *      External declarations
 */
extern "C"
{
#if !defined TBLAYOUTPLUGIN_API
```

```
#define TBLAYOUTPLUGIN_API
#endif

/*
*****
*
*   Main procedure - wo013
*
*****
*/
#ifdef WIN32
TBLAYOUTPLUGIN_API
#endif
    void wo013( const int /* NestedPlateNo */,
                int& /* RawPlateNo */,
                int& /* QuantityNormal */,
                int& /* QuantityMirror */,
                int& /* NumberOfNestedParts */)
    {
    }
}
```

WO014

NAME:

WO014 - AutoNest Interface routine - Get Plate Part Transformation

PURPOSE:

The procedure gets the transformation matrix for the nested part.

DECLARATION:

```
void wo014( const int NestedPlateNo,
            const int NestedPartNo,
            int& PlatePartNo,
            int& Nested,
            double& Dx,
            double& Dy,
            double& Angle,
            int& Mirr);
```

INPUT:

Parameters:

NestedPlateNo	The identification number for the nested plate
NestedPartNo	The identification number for the nested plate part within the current nesting, starting from 1

RESULT:

Parameters:

PlatePartNo	The global identification number for the nested part
Nested	TRUE if the plate part has been nested, otherwise FALSE

The following parameters describe the transformation of the plate part from its local co-ordinate system to the nested plate. The transformation is applied in the following order:

1. rotation
2. reflection
3. translation

Dx Plate part translation
Dy

Angle Plate part rotation

Mirr Plate part mirror code:
0 = Plate part not mirrored.
1 = Plate part mirrored in the x-axis
2 = Plate part mirrored in the y-axis

```
----- */
/*
 *      External declarations
 */
extern "C"
{
#ifdef TBLAYOUTPLUGIN_API
#define TBLAYOUTPLUGIN_API
#endif

/*
*****
*
*   Main procedure - wo014
*
*****
*/
#ifdef WIN32
TBLAYOUTPLUGIN_API
#endif
    void wo014( const int /* NestedPlateNo */,
                const int /* NestedPartNo */,
                int& /* PlatePartNo */,
                int& /*Nested */,
                double& /*Dx */,
                double& /*Dy */,
                double& /*Angle */,
                int& /*Mirr */)
    {
    }
}
```

WO015

NAME:

WO015 - AutoNest Interface routine - Cleanup

PURPOSE:

The procedure performs the necessary cleanup of local interface data structures.

DECLARATION:

void wo015 (void);

INPUT:

RESULT:

None.

```
-----*/
/*
 *      External declarations
 */
extern "C"
{
#ifdef TBLAYOUTPLUGIN_API
#define TBLAYOUTPLUGIN_API
#endif

/*
*****
*
*   Main procedure - wo015
*
*****
*/
#ifdef WIN32
TBLAYOUTPLUGIN_API
#endif
    void wo015 ( void)
    {
    }
}
```

WO016**NAME:**

WO016 - AutoNest Interface routine - Define Maximum number of Qualities

PURPOSE:

The procedure defines the number of base qualities and the maximum number of qualities for one base quality.

DECLARATION:

```
void wo016 ( const int NumberOfBaseQualities,  
             const int MaxNumberOfQualities);
```

INPUT:**Parameters:**

NumberOfBaseQualities The number of base qualities

MaxNumberOfQualities The maximum number of qualities for one base quality

RESULT:

None.

```
-----*/  
/*  
 *      External declarations  
 */  
extern "C"  
{  
#if !defined TBLAYOUTPLUGIN_API  
#define TBLAYOUTPLUGIN_API  
#endif
```



```
/*
*****
*
*   Main procedure - wo016
*
*****
*/
#ifdef WIN32
TBLAYOUTPLUGIN_API
#endif
    void wo016 ( const int /* NumberOfBaseQualities */,
                  const int /* MaxNumberOfQualities */)
    {
    }
```

WO017

NAME:

WO017 - AutoNest Interface routine - Nesting Parameters

PURPOSE:

The procedure defines the parameters for autonest session.

DECLARATION:

```
void wo017 ( const int NestSet,
             const int PlateFill,
             const double MaxScrapSingle,
             const double MaxScrapMirror,
             const double MaxScrapMulti,
             const int TypeOfOptimization,
             const char *NestGps);
```

INPUT:

Parameters:

NestSet	Type of nesting: 1 = normal (nesting based on thickness and quality) 2 = nesting based on assembly 3 = nesting based on production date 4 = nesting based on assembly and production date
PlateFill	Fill type of nesting: 1 = normal (fill plate exclusively with parts defined in NestSet) 2 = Add parts from another set (same thickness and quality, but different assembly or production date)
MaxScrapSingle	Maximum allowed scrap percentage for single part nesting
MaxScrapMirror	Maximum allowed scrap percentage for mirror nesting
MaxScrapMulti	Maximum allowed scrap percentage for multi nesting

TypeOfOptimization The type of optimization to be performed:

0 = no optimization

1 = press down

More activities can be added.

NestGps General Purpose String valid for one autonest session

RESULT:

None.

```
----- */
/*
 *      External declarations
 */

extern "C"
{
#ifdef TBLAYOUTPLUGIN_API
#define TBLAYOUTPLUGIN_API
#endif

/*
*****
*
*   Main procedure - wo017
*
*****
*/
#ifdef WIN32
TBLAYOUTPLUGIN_API
#endif
void wo017 ( const int /* NestSet */,
             const int /* PlateFill */,
             const double /* MaxScrapSingle */,
             const double /* MaxScrapMirror */,
             const double /* MaxScrapMulti */,
             const int /* TypeOfOptimization */,
             const char * /* NestGps */)
{
}
}
```

WO018

NAME:

WO018 - AutoNest Interface routine - Create Partly Filled Nest

PURPOSE:

The procedure creates one partly filled nest for automatic filling with remaining parts. Before calling this procedure raw plates and parts must have been defined.

DECLARATION:

```
void wo018( const int NestedPlateNo,
            const int RawPlateNo,
            const int QuantityNormal,
            const int QuantityMirror,
            const char *Assembly,
            const char *ProductionDate,
            const int NumberOfNestedParts);
```

INPUT:

Parameter:

NestedPlateNo	The identification number for the nested plate
RawPlateNo	The identification number for the used raw plate
QuantityNormal	The number of nestings burned normal in the partly filled nesting
QuantityMirror	The number of nestings burned mirrored in the partly filled nesting
Assembly	Assembly where most of the parts are belonging
ProductionDate	Production date for most of the parts
NumberOfNestedParts	The number of nested parts

RESULT:

Parameters:

None

```
-----*/
/*
 *      External declarations
 */
extern "C"
{
#ifdef TBLAYOUTPLUGIN_API
#define TBLAYOUTPLUGIN_API
#endif

/*
*****
 *
 *   Main procedure - wo018
 *
*****
 */
#ifdef WIN32
TBLAYOUTPLUGIN_API
#endif
    void wo018( const int /* NestedPlateNo */,
                const int /* RawPlateNo */,
                const int /* QuantityNormal */,
                const int /* QuantityMirror */,
                const char * /* Assembly */,
                const char * /* ProductionDate */,
                const int /* NumberOfNestedParts */)
    {
    }
}
```

WO019**NAME:**

WO019 - AutoNest Interface routine - Add Part for Partly Filled Nest

PURPOSE:

The procedure inputs the transformation matrix for the nested part on a partly filled nested plate.

DECLARATION:

```
void wo019( const int NestedPlateNo,
            const int NestedPartNo,
            const int PartNo,
            const double Dx,
            const double Dy,
            const double Angle,
            const int Mirr);
```

INPUT:**Parameters:**

NestedPlateNo	The identification number for the nested plate
NestedPartNo	The identification number for a nested part (local within the nested plate)
PartNo	The identification number for a nested plate part (global)
	The following parameters describe the transformation of the plate part from its local co-ordinate system to the nested plate. The transformation is applied in the following order: 1. rotation 2. reflection 3. translation
Dx Dy	Plate part translation
Angle	Plate part rotation

Mirr Plate part mirror code:
 0 = Plate part not mirrored.
 1 = Plate part mirrored in the x-axis
 2 = Plate part mirrored in the y-axis

RESULT:

Parameters:

None

```
-----*/
/*
 *      External declarations
 */
extern "C"
{
#ifdef TBLAYOUTPLUGIN_API
#define TBLAYOUTPLUGIN_API
#endif

/*
*****
 *
 *   Main procedure - wo019
 *
*****
 */
#ifdef WIN32
TBLAYOUTPLUGIN_API
#endif
    void wo019( const int /* NestedPlateNo */,
                const int /* NestedPartNo */,
                const int /* PartNo */,
                const double /* Dx */,
                const double /* Dy */,
                const double /* Angle */,
                const int /* Mirr */)
    {
    }
}
```

WO020**NAME:**

WO020 - AutoNest Interface routine - Get Number of Remained
Parts

PURPOSE:

The procedure gets the number of unnested parts.

DECLARATION:

int wo020(void);

INPUT:**RESULT:**

The number of unnested parts (different part geometries) is
returned.

```
-----*/
/*
 *      External declarations
 */
extern "C"
{
#ifdef TBLAYOUTPLUGIN_API
#define TBLAYOUTPLUGIN_API
#endif

/*
*****
*
*   Main procedure - wo020
*
*****
*/
#ifdef WIN32
TBLAYOUTPLUGIN_API
#endif
    int wo020 ( void)
    {
        return 0;
    }
}
```


WO021

NAME:

WO021 - AutoNest Interface routine - Get Unnested Parts

PURPOSE:

The procedure gets the parts which were not nested on the basis of the given reason.

DECLARATION:

```
void wo021( const int RemainedPartNo,
            int& PartNo,
            int& QuantityNormal,
            int& QuantityMirror,
            int& ReasonFlag);
```

INPUT:

Parameters:

RemainedPartNo	The identification number for the remained part
----------------	---

RESULT:

Parameters:

PartNo	The identification number for the part
QuantityNormal	The quantity of remained normal image parts
QuantityMirror	The quantity of remained mirrored image parts
ReasonFlag	The reason flag: 0 = unspecified 1 = no raw plate exists for this property 2 = not enough raw plates for this property 3 = the part does not fit on plate (too big)

-----*/

```
/*
 *      External declarations
 */
extern "C"
{
#ifdef TBLAYOUTPLUGIN_API
#define TBLAYOUTPLUGIN_API
#endif

/*
*****
*
*   Main procedure - wo021
*
*****
*/
#ifdef WIN32
TBLAYOUTPLUGIN_API
#endif
    void wo021( const int /* RemainedPartNo */,
                int& /* PartNo */,
                int& /* QuantityNormal */,
                int& /* QuantityMirror */,
                int& /* ReasonFlag */)
    {
    }
}
```

NAME:

W0022 - AutoNest Interface routine - Get Name and Type of
Nested Plate

PURPOSE:

The procedure gets the number of nested parts for the given
nested plate.

DECLARATION:

```
void wo022( const int NestedPlateNo,  
            char* NestingName,  
            int& NestingType,  
            char* DrawingNumber);
```

INPUT:

Parameters:

NestedPlateNo	The identification number for the nested plate
---------------	---

RESULT:

Parameters:

NestingName	The name of the nested plate
NestingType	The type of the nested plate: = 0 Normal, no name returned = 1 NC = 2 FP
DrawingNumber	The drawing number

----- */

```
/*
 *      External declarations
 */
extern "C"
{
#ifdef TBLAYOUTPLUGIN_API
#define TBLAYOUTPLUGIN_API
#endif

/*
*****
*
*   Main procedure - wo022
*
*****
*/
#ifdef WIN32
TBLAYOUTPLUGIN_API
#endif
    void wo022( const int /* NestedPlateNo */,
                char* /* NestingName */,
                int& /* NestingType */,
                char* /* DrawingNumber */)
    {
    }
}
```

W0023**NAME:**

W0023 - AutoNest Interface routine - Define Number of Bevel
Intervals

PURPOSE:

The procedure defines the number of bevel intervals for the
current contour.

DECLARATION:

```
void wo023( const int PlatePartNo,  
            const int ContourNo,  
            const int NumberOfBevels);
```

INPUT:**Parameters:**

PlatePartNo	The identification number for the plate part
ContourNo	The contour number
NumberOfBevels	The number of bevel intervals (only relevant for outer contours and holes).

RESULT:

None.

-----*/

```
/*
 *      External declarations
 */
extern "C"
{
#ifdef TBLAYOUTPLUGIN_API
#define TBLAYOUTPLUGIN_API
#endif

/*
*****
*
*   Main procedure - wo023
*
*****
*/
#ifdef WIN32
TBLAYOUTPLUGIN_API
#endif
    void wo023( const int /* PlatePartNo */,
                const int /* ContourNo */,
                const int /* NumberOfBevels */)
    {
    }
}
```

WO024

NAME:

WO024 - AutoNest Interface routine - Add Bevel Interval for Contour

PURPOSE:

The procedure adds a bevel interval for the current contour.

DECLARATION:

```
void wo024( const int PlatePartNo,
            const int ContourNo,
            const int BevelIntervalNo,
            const int SegPartStart,
            const double StartPoint[2],
            const int SegPartEnd,
            const double EndPoint[2],
            const double BevelCode);
```

INPUT:

Parameters:

PlatePartNo	The identification number for the plate part
ContourNo	The contour number
BevelIntervalNo	The bevel interval number
SegPartStart	Segment part number for the bevel interval start
StartPoint	Interval starting point
SegPartEnd	Segment part number for the bevel interval end
EndPoint	Interval ending point
BevelCode	The bevel code (< 0 indicates other side)

RESULT:

None.

```
----- */
/*
 *      External declarations
 */
extern "C"
{
#ifdef TBLAYOUTPLUGIN_API
#define TBLAYOUTPLUGIN_API
#endif

/*
*****
*
*   Main procedure - wo024
*
*****
*/
#ifdef WIN32
TBLAYOUTPLUGIN_API
#endif
    void wo024( const int /* PlatePartNo */,
                const int /* ContourNo */,
                const int /* BevelIntervalNo */,
                const int /* SegPartStart */,
                const double [2] /* StartPoint */,
                const int /* SegPartEnd */,
                const double [2] /* EndPoint */,
                const double /* BevelCode */)
    {
    }
}
```


WO025**NAME:**

WO025 - AutoNest Interface routine - Get Optimized Raw Plate
Data

PURPOSE:

The procedure gets the data for the selected raw plate.

DECLARATION:

```
void wo025( const int NestedPlateNo,  
            double& Length,  
            double& Width,  
            double& Thickness,  
            char* Quality,  
            double& Density);
```

INPUT:**Parameters:**

NestedPlateNo	The identification number for the nested plate
---------------	---

RESULT:**Parameters:**

Length	The raw plate length
Width	The raw plate length
Thickness	The raw plate length
Quality	The raw plate length
Density	The raw plate length

```
----- */  
/*  
 *      External declarations  
 */  
extern "C"
```

```
{
#if !defined TBLAYOUTPLUGIN_API
#define TBLAYOUTPLUGIN_API
#endif

/*
*****
*
*   Main procedure - wo025
*
*****
*/
#ifdef WIN32
TBLAYOUTPLUGIN_API
#endif
    void wo025( const int /* NestedPlateNo */,
                double& /* Length */,
                double& /* Width */,
                double& /* Thickness */,
                char* /* Quality */,
                double& /* Density */)
    {
    }
}
```

